

Library Management System

Topic : Library Management System

You are going to build a project based on Library Management System. It keeps track of all information about books in the library, their cost, status and total number of books available in the library.

Create a database named library and following TABLES in the database:

1. Branch
2. Employee
3. Books
4. Customer
5. IssueStatus
6. ReturnStatus

Attributes for the tables:

1. Branch

- Branch_no
- Set as PRIMARY KEY
- Manager_Id
- Branch_address
- Contact_no

2. Employee

- Emp_Id – Set as PRIMARY KEY
- Emp_name
- Position
- Salary
- Branch_no
- Set as FOREIGN KEY and it refer Branch_no in Branch table

3. Books

- ISBN
- Set as PRIMARY KEY
- Book_title
- Category
- Rental_Price
- Status [Give yes if book available and no if book not available]

- Author
- Publisher

4. Customer

- Customer_Id
- Set as PRIMARY KEY
- Customer_name
- Customer_address
- Reg_date

5. IssueStatus

- Issue_Id
- Set as PRIMARY KEY
- Issued_cust – Set as FOREIGN KEY and it refer customer_id in CUSTOMER table
- Issued_book_name
- Issue_date
- Isbn_book – Set as FOREIGN KEY and it should refer isbn in BOOKS table

6. ReturnStatus

- Return_Id
- Set as PRIMARY KEY
- Return_cust
- Return_book_name
- Return_date
- Isbn_book2
- Set as FOREIGN KEY and it should refer isbn in BOOKS table

```

1 • CREATE DATABASE library;
2 • USE library;
3
4 • CREATE TABLE Branch (
5     Branch_no INT PRIMARY KEY,
6     Manager_Id INT,
7     Branch_address VARCHAR(100),
8     Contact_no VARCHAR(15)
9 );
10
11 • CREATE TABLE Employee (
12     Emp_Id INT PRIMARY KEY,
13     Emp_name VARCHAR(50),
14     Position VARCHAR(50),
15     Salary DECIMAL(10,2),
16     Branch_no INT,
17     FOREIGN KEY (Branch_no) REFERENCES Branch(Branch_no)
18 );
19
20
21 • CREATE TABLE Books (
22     ISBN VARCHAR(20) PRIMARY KEY,
23     Book_title VARCHAR(100),
24     Category VARCHAR(50),
25     Rental_Price DECIMAL(10,2),
26     Status VARCHAR(3),
27     Author VARCHAR(50),
28     Publisher VARCHAR(50)
29 );
30
31 • CREATE TABLE Customer (
32     Customer_Id INT PRIMARY KEY,
33     Customer_name VARCHAR(50),
34     Customer_address VARCHAR(100),
35     Reg_date DATE
36 );
37
38

```

```

37 • CREATE TABLE IssueStatus (
38     Issue_Id INT PRIMARY KEY,
39     Issued_cust INT,
40     Issued_book_name VARCHAR(100),
41     Issue_date DATE,
42     Isbn_book VARCHAR(20),
43     FOREIGN KEY (Issued_cust) REFERENCES Customer(Customer_Id),
44     FOREIGN KEY (Isbn_book) REFERENCES Books(ISBN)
45 );
46
47 • CREATE TABLE ReturnStatus (
48     Return_Id INT PRIMARY KEY,
49     Return_cust INT,
50     Return_book_name VARCHAR(100),
51     Return_date DATE,
52     Isbn_book2 VARCHAR(20),
53     FOREIGN KEY (Isbn_book2) REFERENCES Books(ISBN)
54 );

```

```

55
56 • INSERT INTO Branch VALUES
57     (1,101,'Kochi','9876543210'),
58     (2,102,'Calicut','9876543211'),
59     (3,103,'Trivandrum','9876543212');
60
61 • INSERT INTO Employee VALUES
62     (101,'Rahul','Manager',60000,1),
63     (102,'Anu','Manager',55000,2),
64     (103,'Vishnu','Manager',70000,3),
65     (104,'Akhil','Clerk',30000,1),
66     (105,'Nisha','Assistant',45000,2),
67     (106,'Arun','Assistant',52000,3);
68
69 • INSERT INTO Books VALUES
70     ('B101','History of India','History',30,'yes','R Sharma','ABC'),
71     ('B102','SQL Basics','Education',20,'yes','John','XYZ'),
72     ('B103','Physics Fundamentals','Science',25,'no','David','PQR'),
73     ('B104','World History','History',35,'yes','Alex','ABC'),
74     ('B105','Python Programming','Education',40,'yes','Mark','XYZ');
75

```

```

6 • INSERT INTO Customer VALUES
7   (1,'Aisha','Malappuram','2021-05-10'),
8   (2,'Fathima','Kozhikode','2023-01-15'),
9   (3,'Rashid','Kochi','2020-11-20'),
0   (4,'Nafla','Thrissur','2022-08-12');
1
2 • INSERT INTO IssueStatus VALUES
3   (1,1,'History of India','2023-06-10','B101'),
4   (2,2,'SQL Basics','2023-06-15','B102'),
5   (3,3,'World History','2023-07-01','B104');
6
7 • INSERT INTO ReturnStatus VALUES
8   (1,1,'History of India','2023-06-20','B101'),
9   (2,2,'SQL Basics','2023-06-25','B102');
0
1 • SHOW TABLES;

```

Result Grid		Filter Rows
	Tables_in_library	
►	books	
	branch	
	customer	
	employee	
	issuestatus	
	returnstatus	

Display all the tables .

92

93 • `SELECT*FROM Branch;`

94

Result Grid

Filter Rows:

Edit:

	Branch_no	Manager_Id	Branch_address	Contact_no
▶	1	101	Kochi	9876543210
	2	102	Calicut	9876543211
	3	103	Trivandrum	9876543212
★	NULL	NULL	NULL	NULL

94

95 • `SELECT*FROM Employee;`

96

Result Grid



Filter Rows:

Edit:


	Emp_Id	Emp_name	Position	Salary	Branch_no
▶	101	Rahul	Manager	60000.00	1
	102	Anu	Manager	55000.00	2
	103	Vishnu	Manager	70000.00	3
	104	Akhil	Clerk	30000.00	1
	105	Nisha	Assistant	45000.00	2
	106	Arun	Assistant	52000.00	3
•	NULL	NULL	NULL	NULL	NULL

Employee 3

97 • `SELECT*FROM Books;`

98

Result Grid				Filter Rows:		Edit:				Export/Import:			V
	ISBN	Book_title	Category	Rental_Price	Status	Author	Publisher						
▶	B101	History of India	History	30.00	yes	R Sharma	ABC						
	B102	SQL Basics	Education	20.00	yes	John	XYZ						
	B103	Physics Fundamentals	Science	25.00	no	David	PQR						
	B104	World History	History	35.00	yes	Alex	ABC						
	B105	Python Programming	Education	40.00	yes	Mark	XYZ						
★	NULL	NULL	NULL	NULL	NULL	NULL	NULL						

98

99 • `SELECT*FROM Customer;`

100

Result Grid	Filter Rows:	Edit:	Export
Customer_Id	Customer_name	Customer_address	Reg_date
1	Aisha	Malappuram	2021-05-10
2	Fathima	Kozhikode	2023-01-15
3	Rashid	Kochi	2020-11-20
4	Nafia	Thrissur	2022-08-12
NULL	NULL	NULL	NULL

11 • **SELECT*FROM IssueStatus;**

Result Grid

Filter Rows:

Edit:

Export

	Issue_Id	Issued_cust	Issued_book_name	Issue_date	Isbn_book
	1	1	History of India	2023-06-10	B101
	2	2	SQL Basics	2023-06-15	B102
	3	3	World History	2023-07-01	B104
	NULL	NULL	NULL	NULL	NULL

13 • **SELECT* FROM ReturnStatus;**

14

Result Grid	Filter Rows:	Edit:	Export/Import:	
Return_Id	Return_cust	Return_book_name	Return_date	Isbn_book2
1	1	History of India	2023-06-20	B101
2	2	SQL Basics	2023-06-25	B102
NULL	NULL	NULL	NULL	NULL

1. Retrieve the book title, category, and rental price of all available books.

```

.05 • SELECT Book_title, Category, Rental_Price
.06 FROM Books
.07 WHERE Status = 'yes';
.08

```

Result Grid  Filter Rows: Export: 			
	Book_title	Category	Rental_Price
▶	History of India	History	30.00
	SQL Basics	Education	20.00
	World History	History	35.00
	Python Programming	Education	40.00

2. List the employee names and their respective salaries in descending order of salary.

```

.09 • SELECT Emp_name , Salary
.10 FROM Employee
.11 ORDER BY Salary DESC;
.12

```

Result Grid  Filter Rows:		
	Emp_name	Salary
▶	Vishnu	70000.00
	Rahul	60000.00
	Anu	55000.00
	Arun	52000.00
	Nisha	45000.00
	Akhil	30000.00

Employee 9 ✕

3. Retrieve the book titles and the corresponding customers who have issued those books.

112

```
113 • SELECT i.Issued_book_name, c.Customer_name
114 FROM IssueStatus i
115 JOIN Customer c
116 ON i.Issued_cust = c.Customer_Id;
```

Result Grid			Filter Rows:
	Issued_book_name	Customer_name	
▶	History of India	Aisha	
	SQL Basics	Fathima	
	World History	Rashid	

4. Display the total count of books in each category.

11

```
8 • SELECT Category, COUNT(*) AS Total_Books
9 FROM Books
10 GROUP BY Category;
```

11

Result Grid			Filter Rows:
	Category	Total_Books	
▶	History	2	
	Education	2	
	Science	1	

5. Retrieve the employee names and their positions for the employees whose salaries are above Rs.50,000.

```

12 • SELECT Emp_name , Position
13 FROM Employee
14 WHERE Salary > 50000;

```

Result Grid			Filter Rows:
	Emp_name	Position	
▶	Rahul	Manager	
	Anu	Manager	
	Vishnu	Manager	
	Arun	Assistant	

6. List the customer names who registered before 2022-01-01 and have not issued any books yet.

```

5
6 • SELECT Customer_name
7 FROM Customer
8 WHERE Reg_date < '2022-01-01'
9 AND Customer_Id NOT IN
10 (
11 SELECT Issued_cust
12 FROM IssueStatus
13 );

```

Result Grid		Filter Rows:
	Customer_name	

7. Display the branch numbers and the total count of employees in each branch.

```

1
2 • SELECT Branch_no, COUNT(*) AS Employee_Count
3 FROM Employee
4 GROUP BY Branch_no;

```

Result Grid			Filter Rows:
	Branch_no	Employee_Count	
▶	1	2	
	2	2	
	3	2	

8. Display the names of customers who have issued books in the month of June 2023.

- ```

SELECT DISTINCT c.Customer_name
FROM Customer c
JOIN IssueStatus i
ON c.Customer_Id = i.Issued_cust
WHERE MONTH (i.Issue_date)=6
AND YEAR (i.Issue_date)=2023;

```

| Result Grid |               | Filter Row |
|-------------|---------------|------------|
|             | Customer_name |            |
| ▶           | Aisha         |            |
|             | Fathima       |            |

9. Retrieve book\_title from book table containing history.

```

SELECT Book_title
FROM Books
WHERE Book_title LIKE '%History%';

```

| Result Grid |                  | Filter Rows: |
|-------------|------------------|--------------|
|             | Book_title       |              |
| ▶           | History of India |              |
|             | World History    |              |

10. Retrieve the branch numbers along with the count of employees for branches having more than 5 employees

```
SELECT Branch_no, COUNT(*) AS Employee_Count
FROM Employee
GROUP BY Branch_no
HAVING COUNT(*) > 5;
```

| Result Grid    Filter Rows: <input type="text" value="E"/> |                |
|---------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Branch_no                                                                                                                                   | Employee_Count |

11. Retrieve the names of employees who manage branches and their respective branch addresses.

```
SELECT e.Emp_name, b.Branch_address
FROM Employee e
JOIN Branch b
ON e.Emp_Id = b.Manager_Id;
```

| Result Grid    Filter Rows: <input type="text"/> |                |
|-------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Emp_name                                                                                                                            | Branch_address |
| Rahul                                                                                                                               | Kochi          |
| Anu                                                                                                                                 | Calicut        |
| Vishnu                                                                                                                              | Trivandrum     |

12. Display the names of customers who have issued books with a rental price higher than Rs. 25.

```
SELECT DISTINCT c.Customer_name
FROM Customer c
JOIN IssueStatus i
ON c.Customer_Id = i.Issued_cust
JOIN Books b
ON i.Isbn_book = b.ISBN
WHERE b.Rental_Price > 25;
```

Result Grid |   Filter Rows:

|   | Customer_name |
|---|---------------|
| ▶ | Aisha         |
|   | Rashid        |